

CLOSURE PROPERTIES OF REGULAR SETS

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LEARNING OBJECTIVE

- To construct finite automata for any given pattern and find its equivalent regular expressions
 - To understand the properties of FA and RE

DEFINITION OF REGULAR EXPRESSION

- Let Σ be an alphabet. The regular expressions over Σ and the sets that they denote are defined recursively as follows:
 1. ϕ is a regular expression and denotes the empty set.
 2. ε is a regular expression and denotes the set $\{\varepsilon\}$
 3. For each $a \in \Sigma$, ' a ' is a regular expression and denotes the set $\{a\}$.
 4. If r and s are regular expressions denoting the languages R and S respectively then $(r + s)$, (rs) , $(r)^*$ are regular expressions that denotes the sets $R \cup S$, RS and R^* respectively.

THEOREM 1

- Regular sets are closed under union, concatenation, and closure
- Proof
 - If L_1 and L_2 are regular, then there are regular expressions r_1 and r_2 denoting the languages L_1 and L_2 , respectively.
 - By definition of RE “If r and s are regular expressions denoting the languages R and S respectively then $(r + s)$, (rs) , $(r)^*$ are regular expressions that denotes the sets $R \cup S$, RS and R^* respectively”
 - Therefore $(r_1 + r_2)$, $(r_1 . r_2)$ and $(r_1)^*$ are regular expressions denoting the languages $L_1 \cup L_2$, $L_1 . L_2$ and L_1^*

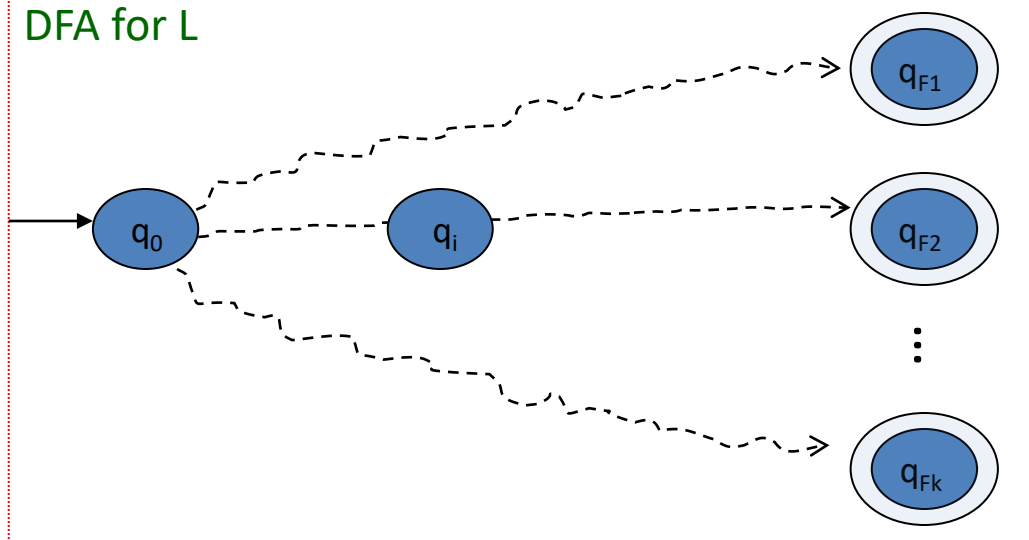
THEOREM 2

- The class of regular sets is closed under complementation. ie If L is a regular set over Σ , then $L^c = \Sigma^* - L$ is a regular set
- Proof
 - Suppose that L is a regular over an alphabet Σ .
 - There is a DFA $M = (Q, \Sigma, \delta, q_0, F)$ accepting L
 - Design a DFA $M' = (Q, \Sigma, \delta, q_0, F')$, where $F' = Q - F$
 - Now, we have that $L(M') = \Sigma^* - L$.
 - Hence, the complement of L is regular

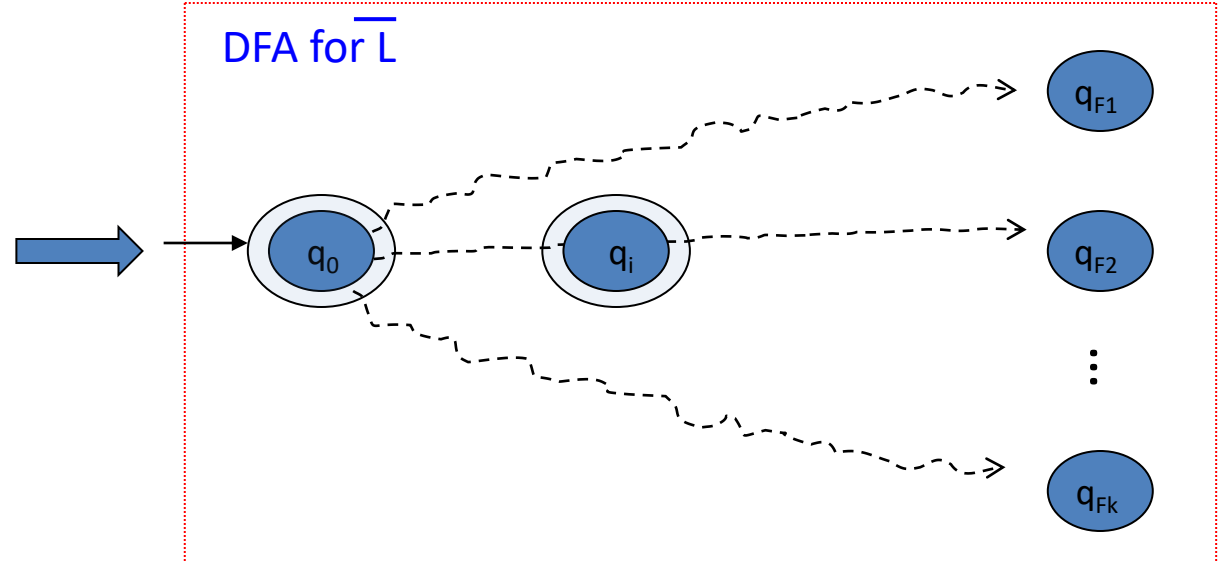
THEOREM 2

Convert every final state into non-final, and every non-final state into a final state

DFA for L



DFA for $\overline{L}$



Assumes q_0 is a non-final state. If q_0 is a final state, invert it.

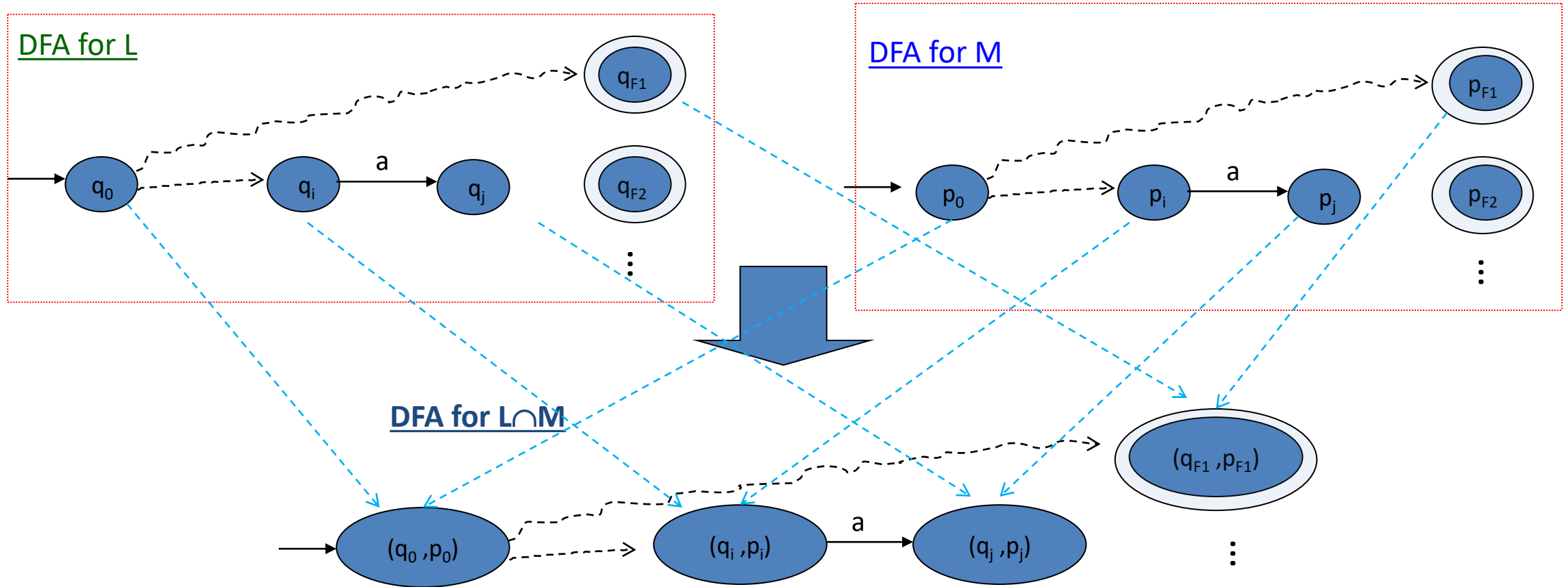
THEOREM 3

- Regular sets are closed under intersection
- Proof
 - By DeMorgan's law:
$$L \cap M = \overline{(\overline{L} \cup \overline{M})}$$
 - Since Regular Sets are closed under union and complementation, they are also closed under intersection

THEOREM 3

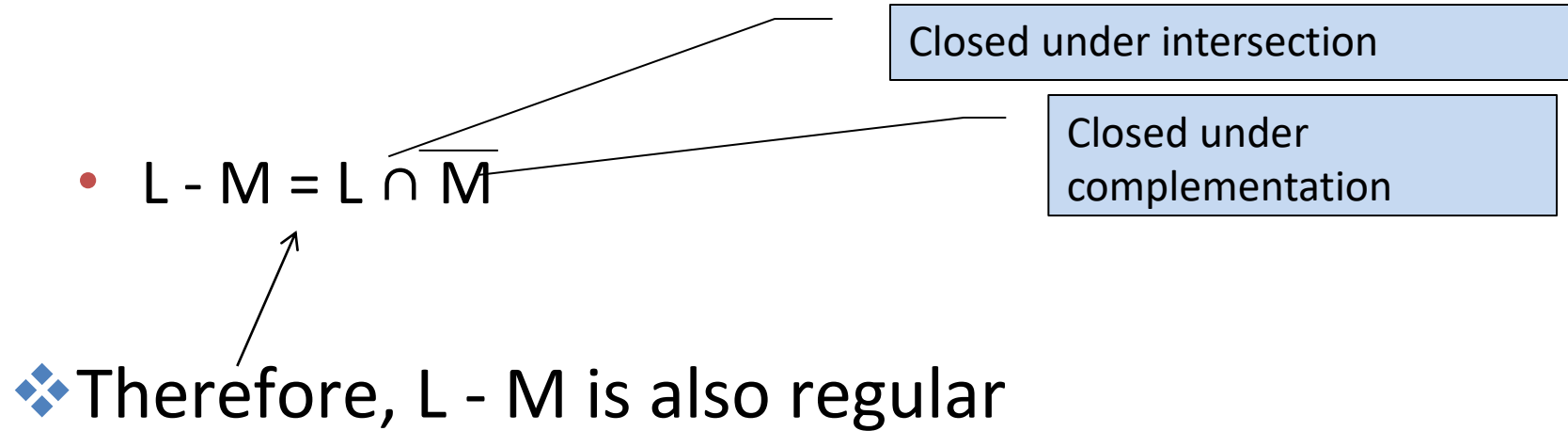
- $A_L = \text{DFA for } L = \{Q_L, \Sigma, q_L, F_L, \delta_L\}$
- $A_M = \text{DFA for } M = \{Q_M, \Sigma, q_M, F_M, \delta_M\}$
- Build $A_{L \cap M} = \{Q_L \times Q_M, \Sigma, (q_L, q_M), F_L \times F_M, \delta\}$ such that:
 - $\delta((p, q), a) = (\delta_L(p, a), \delta_M(q, a))$, where p in Q_L , and q in Q_M
- This construction ensures that a string w will be accepted if and only if w reaches an accepting state in both input DFAs.

THEOREM 3



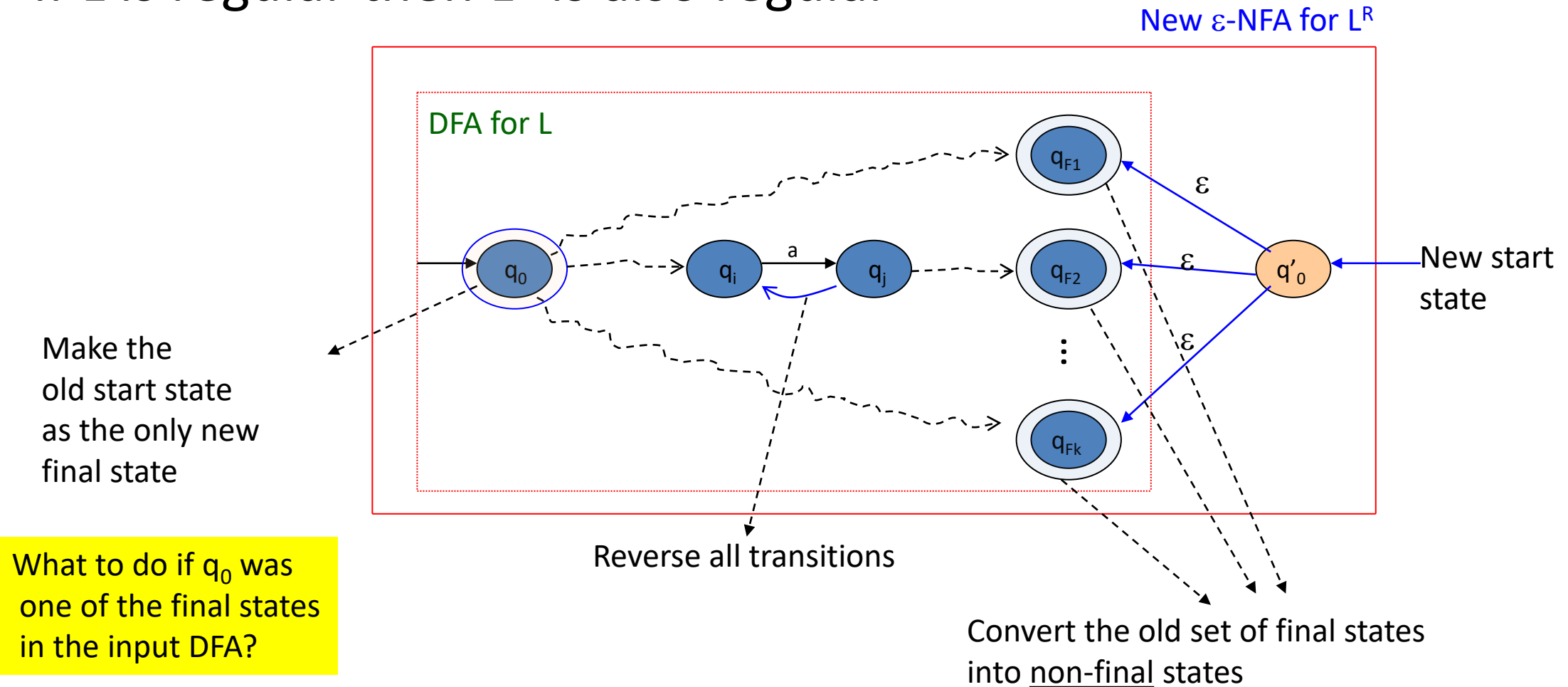
THEOREM 4

- Regular sets are closed under set difference
- Proof



THEOREM 5

- If L is regular then L^R is also regular

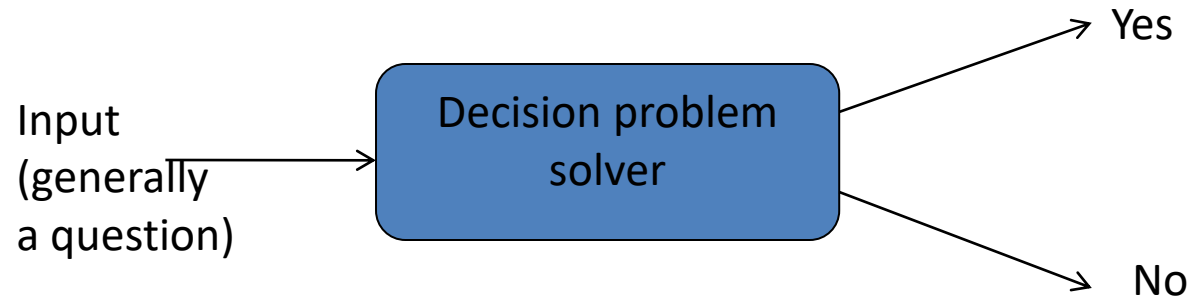


What to do if q_0 was one of the final states in the input DFA?

DECISION PROPERTIES OF REGULAR SETS

INTRODUCTION

- Any “decision problem” looks like this



MEMBERSHIP QUESTION

- Decision Problem: Given L , is w in L ?
- Possible answers: Yes or No
- Approach:
 1. Build a DFA for L
 2. Input w to the DFA
 3. If the DFA ends in an accepting state, then yes; otherwise no.

EMPTINESS TEST

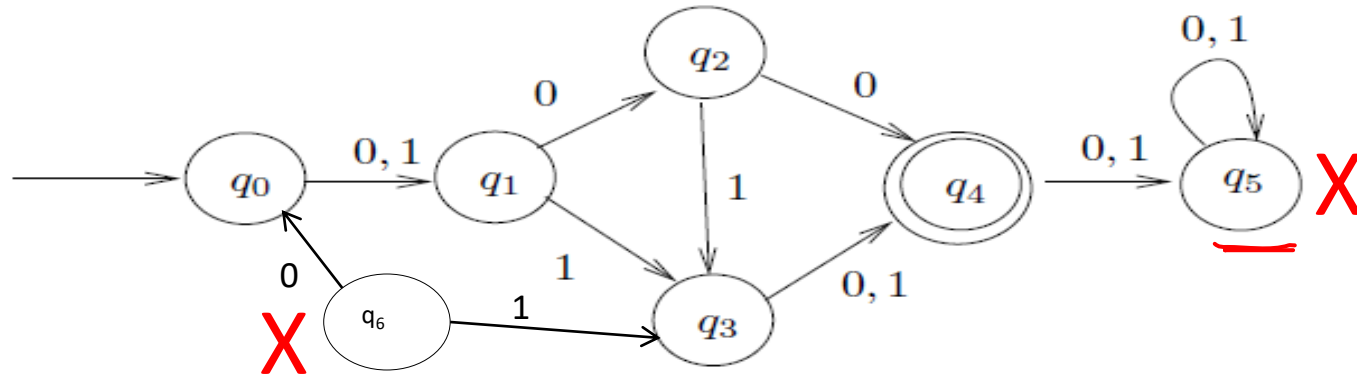
- Decision Problem: Is $L=\emptyset$?
- Approach:
 1. Build a DFA for L
 2. From the start state, run a *reachability* test, which returns:
 1. success: if there is at least one final state that is reachable from the start state
 2. failure: otherwise
 3. $L=\emptyset$ if and only if the reachability test fails

FINITENESS

- Decision Problem: Is L finite or infinite?
- Approach:
 1. Build DFA for L
 2. Remove all states unreachable from the start state
 3. Remove all states that cannot lead to any accepting state.
 4. After removal, check for cycles in the resulting FA
 5. L is finite if there are no cycles; otherwise it is infinite

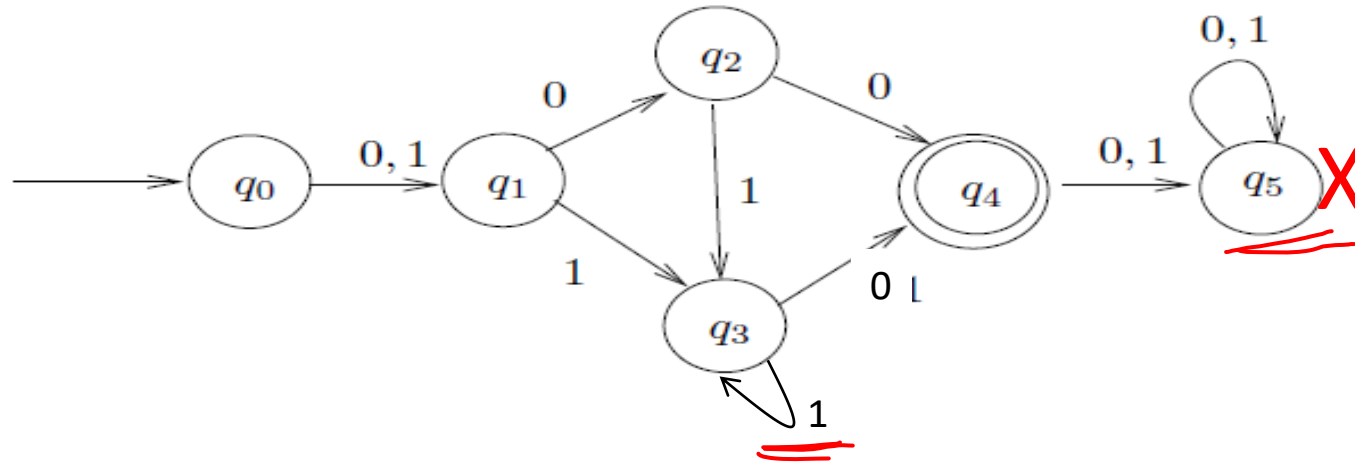
FINITENESS TEST - EXAMPLES

Ex 1) Is the language of this DFA finite or infinite?



FINITE

Ex 2) Is the language of this DFA finite or infinite?



INFINITE

SUMMARY

- Closure Properties of Regular Sets
- Decision Properties of Regular Sets

TEST YOUR KNOWLEDGE

- Regular sets are closed under union, concatenation and kleene closure.
 - a) True
 - b) False
 - c) Depends on regular set
 - d) Can't say
- If L1 and L2 are regular sets then intersection of these two will be
 - a) Regular
 - b) Non Regular
 - c) Recursive
 - d) Non Recursive

TEST YOUR KNOWLEDGE

- Reverse of a DFA can be formed by
 - a) using PDA
 - b) making final state as non-final
 - c) making final as starting state and starting state as final state
 - d) None of the mentioned
- Complement of $(a + b)^*$ will be
 - a) ϕ
 - b) null
 - c) a
 - d) b

LEARNING OUTCOME

On successful completion of this topic, the student will be able to:

- Understand equivalence of FA and RE(K3)

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008